UPI SHIELD AI

Explainable AI-Driven Scam-Pattern Recognition

Team RSG • Muthayammal Engineering College

Hackathon track: AI-Driven Scam Pattern Recognition

Synthetic-data prototype combining transparent rules, Isolation Forest anomaly signals, explainable risk indicators, and a human-review workflow.

SILAMBARASAN C. • RAMANA • GOKUL PRIYA

Prototype only — not connected to real UPI systems and does not initiate or block payments.

1. The problem

Digital payments are fast, but suspicious patterns can be hard to review consistently.

Potential signals include unusually large amounts compared with typical activity, rapid transaction bursts, and new-payee activity. Any one signal can also occur during legitimate use.

The goal is to prioritize review without treating an anomaly as proof of fraud.

2. Our proposed solution

UPI Shield AI combines transparent detection rules with a machine-learning anomaly signal.

- Rule-based checks provide readable reasons for an alert.
- Isolation Forest highlights unusual feature combinations.
- Risk indicators prioritize cases for human review.
- A dashboard and reports support inspection and evaluation.

Design principle: explain the signal, preserve uncertainty, and keep a human in the loop.

3. Prototype workflow

Synthetic transaction events → feature engineering → rule-based detector + Isolation Forest → risk assessment → explanation → analyst review → evaluation report.

Demo inputs include transaction amount, typical amount, recent transaction count, and new-payee status.

No live banking data, UPI integration, payment initiation, or automatic blocking is implemented.

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4. Detection design

Rule-based layer: evaluates predefined patterns such as amount deviation and transaction frequency, then provides interpretable reasons.

ML layer: Isolation Forest identifies unusual feature combinations. Its output is an anomaly signal, not a calibrated probability of fraud.

Hybrid review: combines rule evidence and ML signal to prioritize analyst attention. New-payee status alone should not produce a high-risk conclusion.

Real deployment would require authorized representative data, independent validation, monitoring, and security review.

5. Synthetic evaluation results

Held-out synthetic test set: 90 events · 76 actual normal and 14 actual suspicious.

Metric	Rule-based	Isolation Forest
Precision	100.0%	73.7%
Recall	100.0%	100.0%
F1-score	100.0%	84.8%

Isolation Forest confusion matrix: TN = 71, FP = 5, FN = 0, TP = 14.

These results apply only to this synthetic test set. The rule-based 100% scores may reflect how the synthetic labels and rules were constructed; they do not establish real-world fraud detection performance.

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6. Hybrid scenario validation

Observed scenario-test results: all available hybrid checks passed in the current notebook run.

- Normal activity: score-range check passed; LOW risk returned.
- New payee only: score-range check passed; rule risk remained below HIGH.
- High amount + transaction burst: score-range check passed; HIGH risk returned.

These are targeted scenario checks, not comprehensive security testing. Expand coverage with edge cases, malformed inputs, boundary values, and regression tests.

7. What the metrics mean

- Isolation Forest flagged all 14 synthetic suspicious events but also flagged 5 normal events.
- Precision of 73.7% means some flagged events were false alarms in this evaluation.
- Rule-based metrics are perfect on this particular synthetic set; independent and more realistic testing is necessary.
- Future work: improve scenario diversity, tune thresholds on validation data, test robustness, and check for data leakage.

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8. Privacy, security and safety

- Synthetic demonstration data only; do not enter payment PINs, passwords, or real customer records.
- No payment initiation, verification, or automatic blocking.
- An anomaly is not proof of fraud; human review is required.

Before deployment: data authorization and minimization, access controls, secure audit logs, abuse testing, drift monitoring, and independent evaluation.

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9. Team RSG

SILAMBARASAN C. — suggested role: Team Lead & AI/ML (dataset, detection logic, model evaluation, coordination).

RAMANA — suggested role: Prototype & Dashboard (interface, risk display, explanations, review queue).

GOKUL PRIYA — suggested role: Research, Security & Presentation (threat model, test evidence, documentation, demo preparation).

These are proposed assignments. Confirm and edit them to match each member's actual contribution before submission.

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10. Next steps

1. Re-run all notebook cells from top to bottom and confirm reproducible outputs.
2. Add edge-case and regression tests; investigate any failed test.
3. Capture genuine dashboard screenshots for normal and suspicious scenarios.
4. Document synthetic data generation, train/test split, and limitations.
5. Follow the official hackathon submission rules, format, and deadline.

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Closing

Explainable alerts for better-informed human review.

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Synthetic-data prototype. Results are preliminary and do not represent real-world UPI fraud detection performance.